

wav2VOT

Automatic estimation of voice onset time, closure duration, and burst realisation with wav2vec2

James Tanner¹, Morgan Sonderegger², Jane Stuart-Smith¹, Tyler Kendall³ & Jeff Mielke⁴
¹University of Glasgow ²McGill University ³University of Oregon ⁴North Carolina State University
Interspeech 2026, Sydney, Australia, 27th September-1st October 2026



Centre for
Data Science & AI

Background

- Automatic approaches to data segmentation and annotation now common in most phonetic research pipelines^[1,2]
- Automatic transcription, forced alignment etc
- Many phonetic projects also interested in **sub-phonemic** information, such as **voice onset time (VOT)**
- Current tools for VOT annotation^[3,4] are powerful, but also have limitations:
 - They often require training data (drawn from the target data) to perform well
 - They only predict VOT and not other information about stops (e.g. closure duration)
 - They assume that a stop is fully realised with a burst
- Research has demonstrated potential of large pre-trained speech models (e.g. wav2vec2^[5]) for producing high-quality phonetic annotations with minimal training data

Research Questions

- Can wav2vec2 be utilised for automatic annotation of VOT, closure duration, and burst realisation?
- How well can a trained wav2vec2 model generalise to different kinds of speech data?

Architecture & design

- wav2vec2 consists of feature encoder (FE) block and transformer block
- Standard wav2vec2 FE = 7 1-D convolutional blocks
 - 1 frame = ~25ms of audio
- wav2VOT FE = 4 blocks
 - 1 frame = ~1ms
- Framewise classification head
 - Predicts {window, closure, VOT, lenition} label for each 1ms frame
- Training loss:
 - Framewise cross-entropy loss
 - CTC on overall sequence (e.g. window-closure-VOT-window)

Audio converted to ~1ms frames

Each frame assigned value [0-3]:

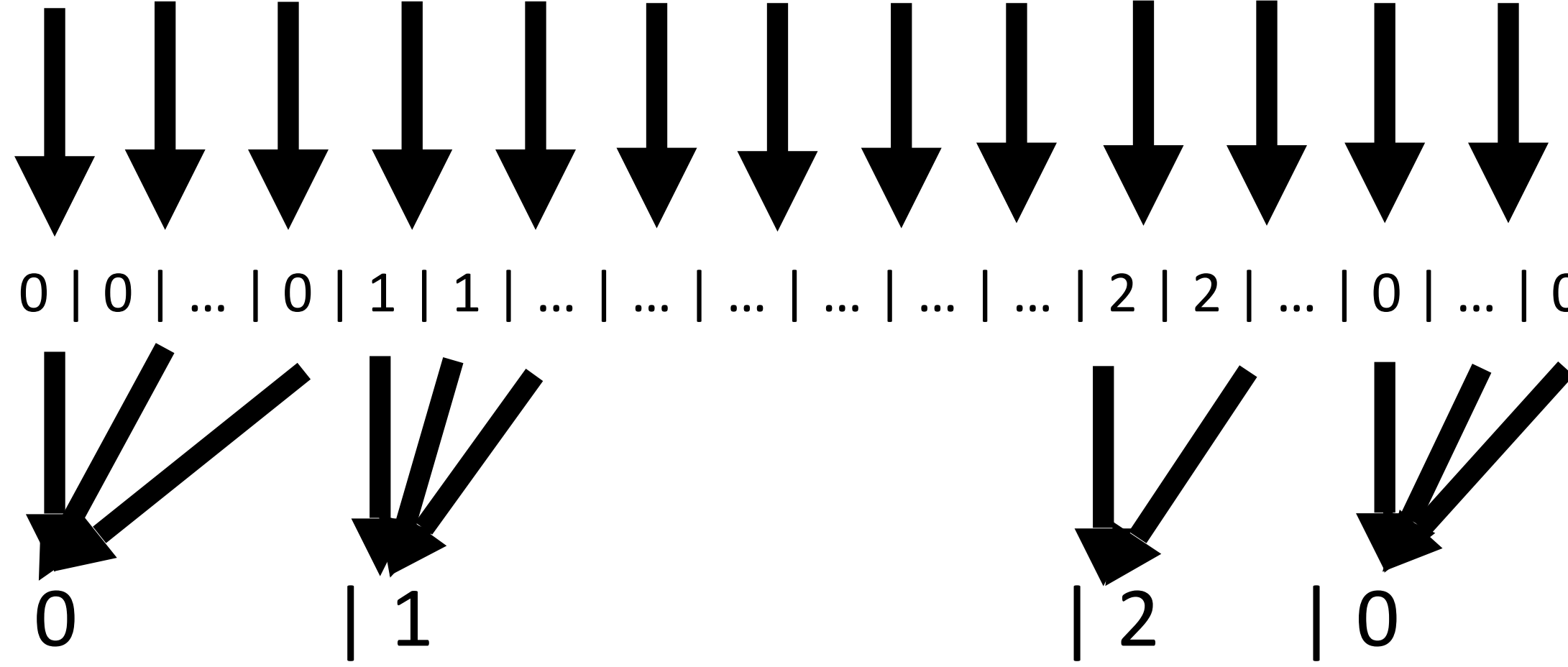
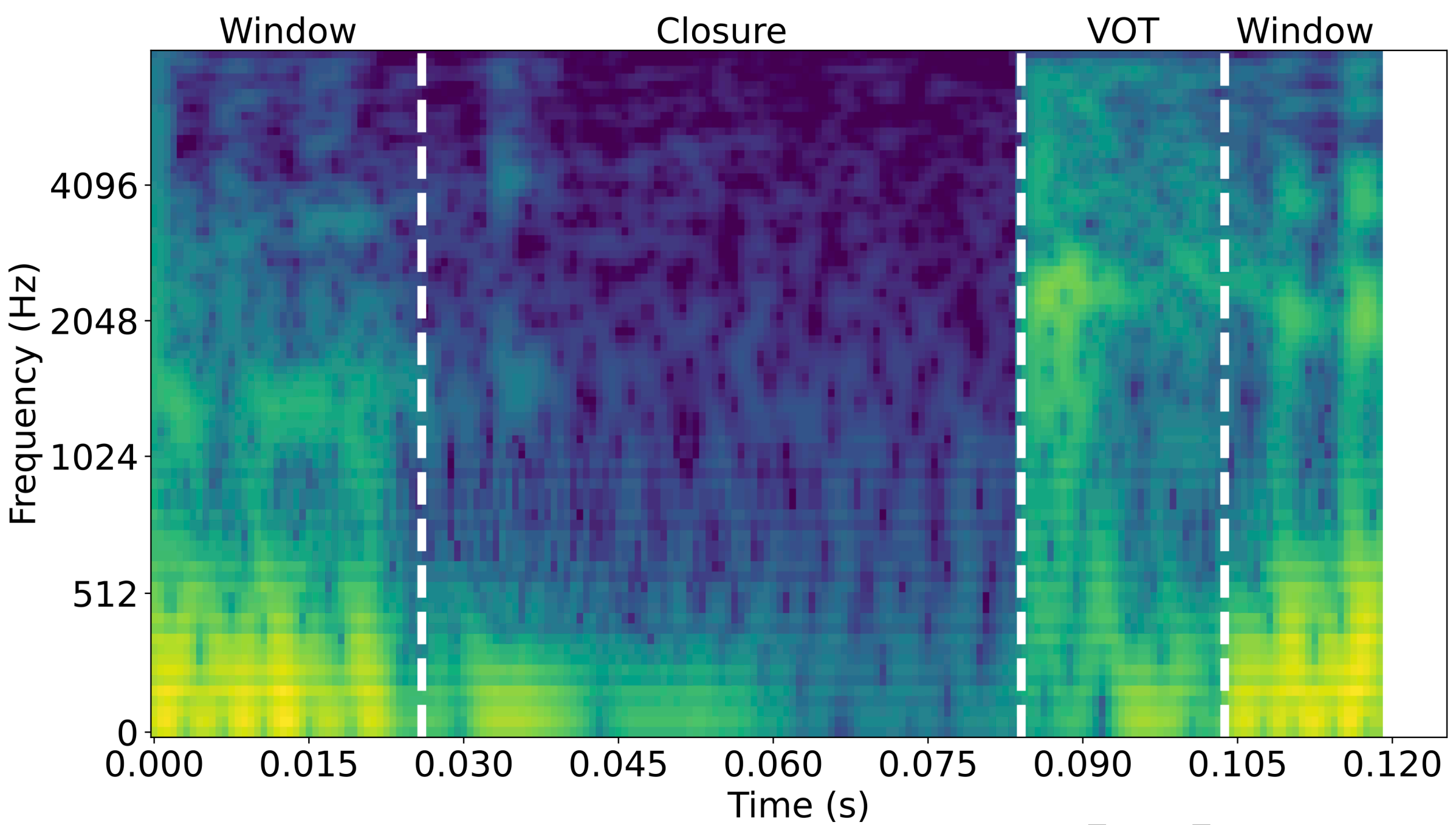
0 = context window

1 = closure

2 = VOT

3 = lenition

Adjacent labels are collapsed



Converted into labelled timestamps

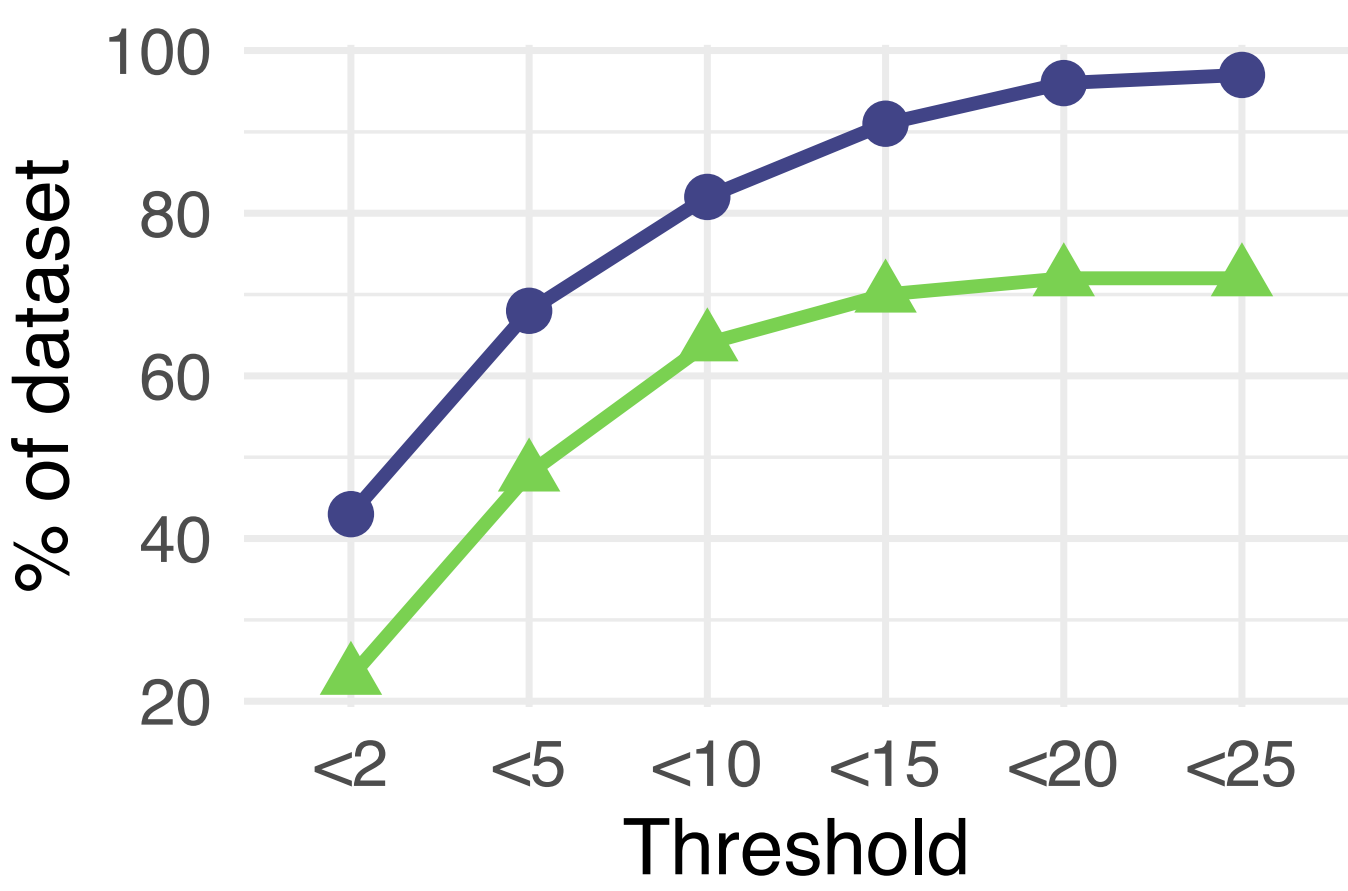
Window | Closure

| VOT | Window

Experiment 1

Test 1

- Train and validate wav2VOT performance using hand-annotated data from spontaneous Japanese speech corpus (CSJ-C^[6], 205k tokens)
- Trained (on 80% training set) for 10 epochs (batch size = 64) on 80GB NVIDIA H100 & validated on 20% holdout set
- Best performing model:
 - 96% framewise accuracy; 0.93 framewise F1



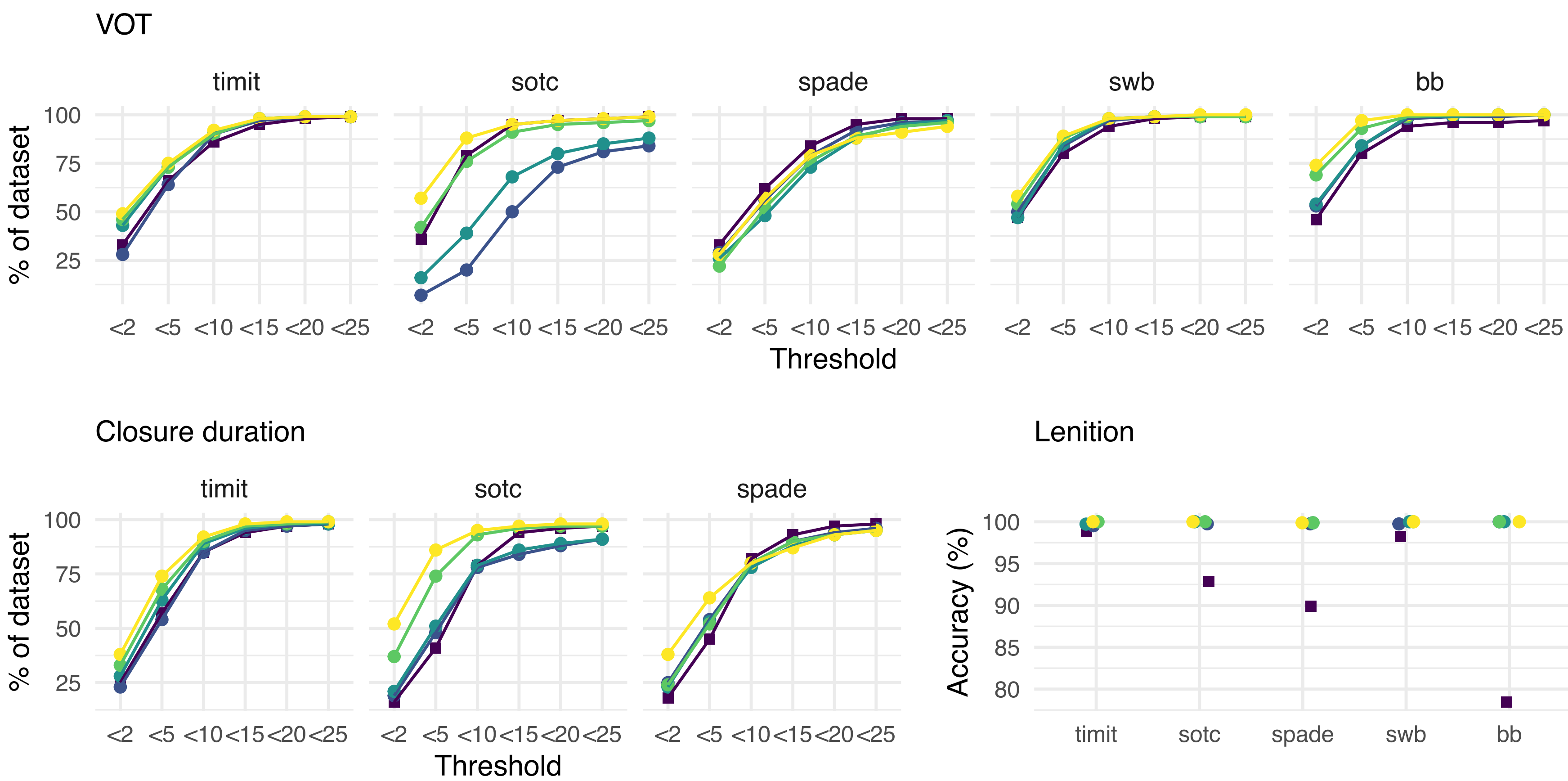
● VOT ▲ Closure duration

Performance of CSJ-C-trained wav2VOT model on CSJ-C holdout set (44k tokens), shown as a percentage of data predicted within a tolerance threshold (e.g. <2 = the proportion of tokens predicted within 2ms of the annotations)

- CSJ-C model performs comparably with other VOT estimation tools (e.g. AutoVOT)
- For SOTC, fine-tuning with 50-100 tokens damages performance (improves with 200+)
- Fine-tuning improves detection of stops with no observed burst

Test 2

- Test how well a wav2VOT model can generalise to different data (e.g. across languages, recording conditions, etc)
- For 5 English speech corpora (with VOT & closure duration annotations):
 - Take 500 training set, fine-tune CSJ-C model each with {50, 100, 200, 500} tokens, & validate on remainder of tokens



Corpora: sotc = Sounds of the City (Glasgow English); spade (British & NA Englishes); swb = Switchboard (NA English); bb = Big Brother (British English)

Training data ● CSJ ● 50 ● 100 ● 200 ● 500

Performance of wav2VOT (top) and closure duration (bottom-left) predictions, reported as the proportion of data that falls within fixed tolerance thresholds. Burst realisation predictions (bottom-right) reported as predictive accuracy. Purple squares indicate CSJ-C model (non-fine-tuned), with circles indicating varying amounts of fine-tuning data used from each test corpus. Closure duration metrics calculated for corpora containing hand-annotated closures (timit, sotc, spade).

References

- [1] Coats, S. (2020). 'Articulation rate in American English in a corpus of YouTube videos', *Language and Speech*, 53, 799-831.
- [2] Brand, J., Hay, J., Clark, L., K. Watson, M. Sósuthy (2021). 'Systematic co-variation of monophthongs across speakers of New Zealand English', *Journal of Phonetics*, 88.
- [3] Sonderegger, M., Keshet, J. (2012). 'Automatic measurement of voice onset time using discriminative structured predictions', *Journal of the Acoustical Society of America*, 132, 3965-3979.
- [4] Shrem, Y., Goldrick, M., Keshet, J. (2019). 'Dr. VOT: measuring positive and negative voice onset time in the wild', *Proc. Interspeech 2019*, 629-633.
- [5] Baevski, A., Zhou, H., Mohamed, A., Auli, M. (2020). 'wav2vec2.0: a framework for self-supervised learning of speech representations', *arXiv:2006.11477*.
- [6] Maekawa, K., Koiso, H., Furui, S., Ishihara, H. (2000). 'Spontaneous speech corpus of Japanese', *Proc. IREC*, 2, 946-952.
- [7] Bürkner, P. C. (2018). 'Advanced Bayesian multilevel modeling with the R package brms', *The R Journal*, 10, 395-411.
- [8] Lenth, R. V. (2024). 'emmeans: estimated marginal means, aka least-squares means'. Package version 1.10.

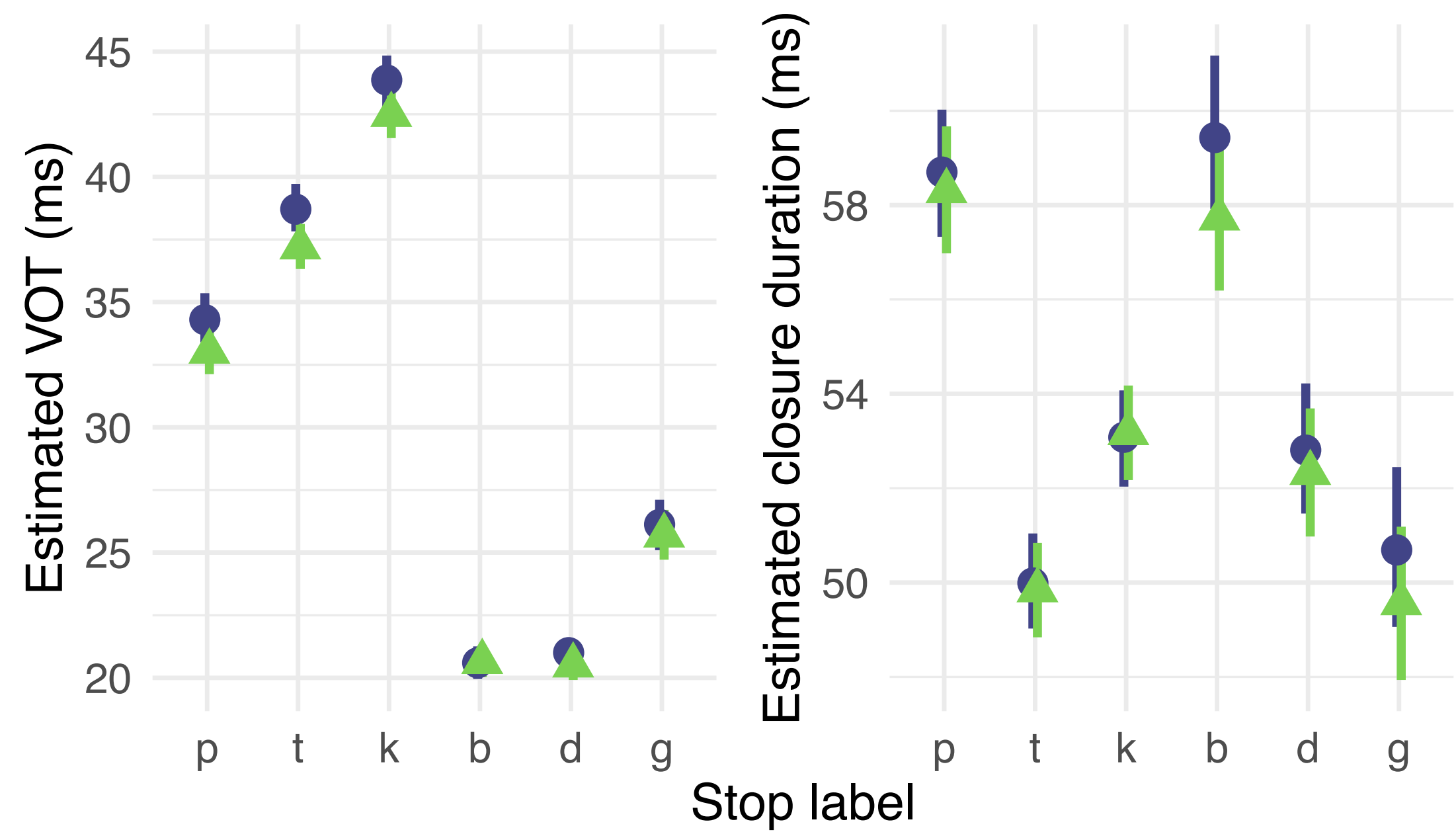
Scan to access
the repository!

<https://github.com/james-tanner/wav2VOT>



Experiment 2

- Compare wav2VOT predictions with hand-annotations in hypothetical phonetic study
- Jointly model VOT and closure duration with Bayesian regression^[7].
 - Fixed terms: source (wav2VOT vs annotation), voicing, POA, speech rate, interactions with source
 - Random terms: speaker intercepts + slopes for all terms, correlation between VOT and closure duration
- No meaningful difference observed between overall wav2VOT predictions and manual annotations
- Evidence for a negligible difference in VOT voicing contrast size (10.7ms vs 10.5ms)
- No differences observed between annotated and predicted closure durations



● Annotation ▲ wav2VOT

Model-estimated VOT (left) and closure duration (right) for manual annotations (purple circles) and wav2VOT predictions (green triangles), separated by phoneme label. Points and lines indicate posterior medians with 95% highest-posterior-density intervals^[8].

Summary

- wav2vec2 can be used to predict VOT, closure duration, and burst realisation
- wav2VOT performs comparably with other VOT estimation tools
- Performance improves with fine-tuning using small training set (~50 tokens)



Digital Research
Alliance of Canada

SPADE
Speech Across Dialects of English

